

Week 3 Recitation

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ECON2001.01 (Prin. Micro.)

Jan 24, 2025

House-keeping

Quiz day!

- Simple comprehension checks, multiple choice
- 15min at the end of the recitation

Exam 1 Next week!

- Jan 28th (Tuesday), 9:10 - 10:05 AM, Independence Hall 100
- 25 Questions; Multiple choice type; Chap 1, 3, 4
 - Lecture slides, Recitation slides, Practice with Textbook Problems and Problem sets!
- We will use the Scantron for the answer sheet.

House-keeping

Scantron Instruction

Last name first! → **Don't forget to bubble the "blank" in between!**

NAME (Last, First, MI) **D O E J O H N**

SEX ☐ M ☐ F

GRADE OR EDUC ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ 21 ☐ 22 ☐ 23 ☐ 24 ☐ 25 ☐ 26 ☐ 27 ☐ 28 ☐ 29 ☐ 30 ☐ 31 ☐ 32 ☐ 33 ☐ 34 ☐ 35 ☐ 36 ☐ 37 ☐ 38 ☐ 39 ☐ 40 ☐ 41 ☐ 42 ☐ 43 ☐ 44 ☐ 45 ☐ 46 ☐ 47 ☐ 48 ☐ 49 ☐ 50 ☐ 51 ☐ 52 ☐ 53 ☐ 54 ☐ 55 ☐ 56 ☐ 57 ☐ 58 ☐ 59 ☐ 60 ☐ 61 ☐ 62 ☐ 63 ☐ 64 ☐ 65 ☐ 66 ☐ 67 ☐ 68 ☐ 69 ☐ 70 ☐ 71 ☐ 72 ☐ 73 ☐ 74 ☐ 75 ☐ 76 ☐ 77 ☐ 78 ☐ 79 ☐ 80 ☐ 81 ☐ 82 ☐ 83 ☐ 84 ☐ 85 ☐ 86 ☐ 87 ☐ 88 ☐ 89 ☐ 90 ☐ 91 ☐ 92 ☐ 93 ☐ 94 ☐ 95 ☐ 96 ☐ 97 ☐ 98 ☐ 99 ☐ 100

Use pencils, not ink pens!

GENERAL PURPOSE - SCANTRON® - ANSWER SHEET
SEE IMPORTANT MARKING INSTRUCTIONS ON SIDE 2

The code assigned to your recitation section - See the exam cover!

BIRTHDATE MO. DAY YR. **IDENTIFICATION NUMBER** A B C D E F G H I J K L M N O P **SPECIAL CODES**

Jan. ☐ Feb. ☐ Mar. ☐ Apr. ☐ May ☐ Jun. ☐ Jul. ☐ Aug. ☐ Sep. ☐ Oct. ☐ Nov. ☐ Dec. ☐

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Review

Revisiting **Marginal** analysis

Review of “**Marginals**” last class!

Review

Equilibrium

We look at **Perfectly Competitive Market**

- Identical goods or services (e.g., agriculture goods)
- Large number of buyers and sellers
 - ⇒ They are **Price-takers!**
 - (No power to impact the market price)

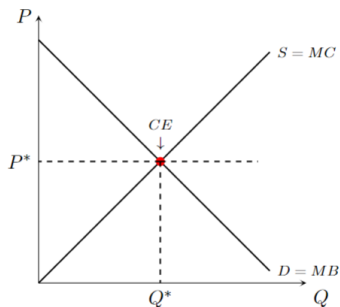
Review

Equilibrium

- Buyer (Consumer, Demand-er)
 - **Demand**: The relationship b/t Price and Q-demanded
 - **Demand schedule** – a table of Price and Q-demanded (Q_d, P)
How to read?
 - **Demand Curve** – Plots the Q-demanded at each price. (Q_d, P)
 - **Willingness to Pay** = the height of the demand curve at each Q
= the buyer's **Marginal Benefit**!
- Seller (Producer, Supplier)
 - **Supply**: The relationship b/t Price and Q-supplied
 - **Supply schedule** – a table of Price and Q-supplied (Q_s, P)
How to read?
 - **Supply Curve** – Plots the Q-supplied at each price. (Q_s, P)
 - **Willingness to Accept** = the height of the supply curve at each Q
= the Seller's **Marginal Cost**!
- Plotting the market demand/supply schedules – Demand/Supply curves

Review

Equilibrium



Recall:

- Equilibrium: the state where all agents simultaneously optimize and that no agent would be better off deviating from the behavior
- Optimum occurs: $MB = MC$

Competitive Equilibrium (CE)

- Optimal price and Optimal quantity
- The unique price, P^* , that equates Q -demanded and Q -supplied, i.e., the intersection of Supply and Demand: $MB = MC$!!

Review

Change in Quantity Demanded vs. Change in Demand

Change in **Q-demanded**

- Caused by – **Price** change!
- Graphically – Movement **along** the curve

Change in **Demand**

- Caused by – **Non-price** change!
 - Tastes and Preferences,
 - Income or wealth,
 - Prices of OTHER goods,
 - Number and scale of buyers (Population changes)
 - Buyers' prospect about the economic condition in the future
- Graphically – shift of the **entire** curve

Review

Change in Quantity Supplied vs. Change in Supply

Change in **Q-Supplied**

- Caused by – **Price** change!
- Graphically – Movement **along** the curve

Change in **Supply**

- Caused by – **Non-price** change!
 - Input price (Labor, Capital, Land(rent))
 - Technology (Ability/Knowledge to produce output)
 - Number and scale of firms (Suppliers)
 - Sellers' prospect about the economic condition in the future
- Graphically – shift of the **entire** curve

Review

Changes in Equilibrium

Analyzing Changes in Equilibrium

Steps:

- 1 Decide whether the event affects the demand side, supply side, or, in some cases, both sides
- 2 Decide whether the curve(s) shifts, and if so, to the right or to the left
- 3 Use the Demand-Supply diagram
 - Compare the initial and the new equilibrium
 - Determine the effects on equilibrium price and quantity

Review

Example

Let's look at the Umbrella market.

Suppose the weather is unusually wet this spring, and everyone needs more umbrellas.

- ① *Step 1:* Demand side
- ② *Step 2:* D curve shifts to the right. (At any given price, increased Q-demanded)
- ③ *Step 3:* When D curve goes to the right, Eqm P $\uparrow$, Eqm Q $\uparrow$

Suppose an unexpected flooding damaged the umbrella firms.

- ① *Step 1:* Supply side
- ② *Step 2:* S curve shifts to the left. (At any given price, decreased Q-supplied)
- ③ *Step 3:* When S curve goes to the left, Eqm P $\uparrow$, Eqm Q $\downarrow$

Review

Example

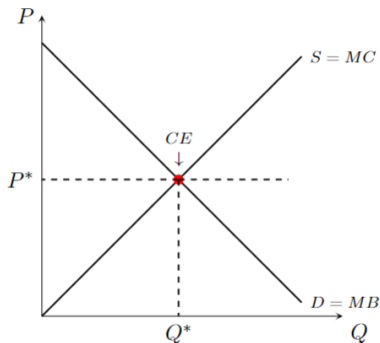
Suppose the weather is unusually wet this spring, but a new technology to produce umbrellas has been developed.

- ① *Step 1:* Demand and Supply side
- ② *Step 2:* D curve shifts to the right; S curve shifts to the right.
(At any given price, reduced cost to produce umbrellas)
- ③ *Step 3:* No matter what, the Eqm Q will increase. However, whether the Eqm P increases or decreases depends on the relative size of each shift.

Review

When price is not equal to the CE price

Excess Supply / Excess Demand



Example: (Textbook Practice Problem #10.)

New York decides to reduce the consumption of sugary soda by imposing a minimum price of \$2.50 per soda. The current equilibrium price is \$1.50. Sketch the supply and demand curves for soda and show the effect of this policy.

Problem Set Exercise

Chapter 4

1. Answer the following questions regarding the fictional Varmint Whiskey market. Assume the market is perfectly competitive. Meaning, all distillers produce the same exact whiskey—no variation in flavor, coloring, bottling, or branding—and no one economic agent can impact the market through their choices.
 - (a) Assume for ease of analysis that our market for Varmint Whiskey consist of two buyers, Wynonna and Henry. Their demand for Vermint Whiskey is represented by the below inverse demand function.

$$P_i = 42 - 4Q_i \quad i = \text{Wynonna, Henry}$$

Using the inverse demand function given above, complete the following table.

Table 1: Demand Schedule

P	Q_{Wynonna}	Q_{Henry}	Q_{Market}
42			
40			
38			
16			
4			
2			
0			

Problem Set Exercise

Chapter 4

- (c) The purpose of this question, as well as part (d), is to increase your understanding of how to interpret demand in various presentations (demand schedule, function, graphically).

Refer to the values that you calculated for part (a) and derive the market inverse demand function. (Hint, the demand function takes the form $P_{\text{market}} = 42 + b \cdot Q_{\text{market}}$ and $P(Q)$ is linear in Q . You need to derive the value of b .)

- (d) Refer to your answer in part (c). For every additional bottle of Varmint Whiskey consumed, how does marginal benefit change? Does your answer satisfy the Law of Demand? Explain your answer.

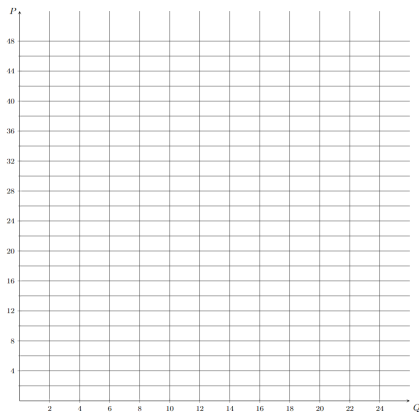
Problem Set - Exercise

Chapter 4

- (c) Suppose the market supply of Varmint Whiskey is characterized by the following inverse supply function,

$$P = 1.5Q \quad (1)$$

In the below space, graph the market supply and demand curves (market demand was derived in part (c)). Clearly label all curves.



Problem Set - Exercise

Chapter 4

- (f) Suppose the per bottle price of Varmint Whiskey is \$12. At that price, is there excess supply or demand, and if so, what is the amount of surplus or shortage that occurs? (Hint, this questions can be answered using the graph or the supply and demand functions)

- (g) Refer to Figure 1. What is the competitive equilibrium (CE)? In addition to stating the CE, indicate it graphically using a small circle and labeling it “E” (for equilibrium) on the above graph.

- (i) Suppose a long and harsh winter impacts both consumers and distillers of Varmint Whiskey. The abnormally low temperatures resulted in below normal rye yields (in ingredient of whiskey). Also, the long winter has resulted in people finding more solace in having a glass of whiskey to lift their spirits and warm the soul.

How will the long and harsh winter affect the competitive equilibrium? Explicitly state how the equilibrium price and quantity will change as a result of the winter.

Problem Set - Exercise

Chapter 4

- (j) The following information reports the specific impact of the harsh winter described in part (i). It is revealed that the cost of producing an additional bottle of whiskey increased by \$10. Also, the comfort that whiskey drinkers found during the long winter amounts to an additional benefit of \$17 per bottle.

Apply this new information to the supply (equation 1) and demand functions (part (c)) and with the slope term derived in part (c)) and solve for the new competitive equilibrium. Make sure to clearly state the new equilibrium price and quantity.